

Vectors, Geometry & Similarity

Mathematics Foundations — Session 1

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Today's Goal

- Rebuild vector intuition in $\mathbb{R}^n$: addition, scalar multiplication, 2D/3D
 - Read the dot product as a measure of alignment between two vectors
 - Use cosine similarity to compare real data
- Deliverable: rank the most similar items in a real dataset using cosine similarity

Plan

1. Vectors in $\mathbb{R}^n$
2. Dot Product
3. First Set of Exercises
4. Angle, Cosine Similarity, Norm & Distance
5. Python Example (NumPy)
6. Final Exercises
7. Wrap-Up

Vectors in $\mathbb{R}^n$

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Vectors: What Are They?

Vectors: Example and Elements of a Vector

What Can We Store in a Vector?

Vector Properties

Vector Properties: Geometrical Meaning

Vector Decomposition

Vector Independence

Notion of Basis

Dot Product

1. Vectors in $\mathbb{R}^n$
- 2. Dot Product**
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Dot Product Operation

Euclidean Norm

Orthogonal and Orthonormal Basis

Norms (General Case)

First Set of Exercises

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Exercises (1/2)

Angle, Cosine Similarity, Norm & Distance

1. Vectors in $\mathbb{R}^n$
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- 4. Angle, Cosine Similarity, Norm & Distance**
5. Python Example (NumPy)
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Angle, Cosine Similarity, Norm & Distance

Link with ML

Cosine and Similarity (1/2)

Cosine and Similarity (2/2)

Link Between the Dot Product, Norm and Distance

Link with LLMs

Python Example (NumPy)

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Python Example Using NumPy

Final Exercises

1. Vectors in $\mathbb{R}^n$
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Exercises (2/2)

Wrap-Up

1. Vectors in $\mathbb{R}^n$
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Recap

To Do for Next Session

Class Feedback

Appendix

8. Appendix

Coordinate Grid

